

Project P2: Windows Security Monitoring & Threat Hunting

Windows 11 Endpoint Auditing, Sysmon Telemetry Engineering, Kali Attacker Investigation & Splunk Dashboards

STATUS: COMPLETE & VERIFIED

SIEM: SPLUNK ENTERPRISE 10.4.3

AGENT: SPLUNK UF 10.4.3

TELEMETRY: SYSMON + WINEVENTLOG

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Date:	September 11, 2026	Classification:	TLP:CLEAR / Lab Deliverable
Target Endpoint:	Windows 11 Pro (192.168.100.8)	Attacker Node:	Kali Linux (192.168.100.6)

1. Executive Summary

This report details the architectural deployment, telemetry pipeline engineering, detection rules, and incident investigation deliverables for **Project P2: Windows Security Monitoring with Splunk**. Operating in an isolated virtual laboratory network (192.168.100.0/24), an enterprise endpoint monitoring environment was constructed on a **Windows 11** workstation (DESKTOP-US2NJE2).

Using the **Splunk Universal Forwarder (UF 10.4.3)** and **Microsoft Sysinternals Sysmon**, rich endpoint telemetry—including process executions (Event ID 1), network connections (Event ID 3), DNS queries (Event ID 22), and Windows Security event logs (Logons, PAM, Object Access)—was continuously ingested over TCP port 9997 into dedicated `windows` and `sysmon` indexes on **Splunk Enterprise**.

TOTAL SECURITY EVENTS	SYSMON EVENTS	PROCESS CREATIONS	ATTACKER FAILED LOGONS
8,638	8,406	3,994	4625
Windows Security logs	Process & Network logs	Sysmon Event ID 1	From Kali 192.168.100.6

The project developed two operational dashboards: the **SOC Threat Hunting Dashboard** and the **Kali Attacker Activity Dashboard**. Controlled reconnaissance and SMB interaction from an adversary node (192.168.100.6) verified 100% detection efficacy for network authentication failures (Event ID 4625) and session correlations (Event ID 4624).

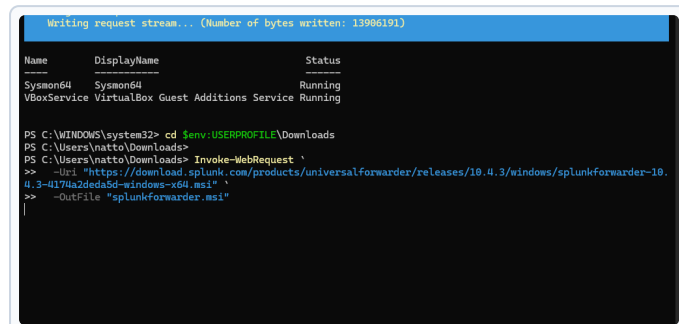
2. Lab Environment & System Architecture

The lab environment is connected via Oracle VirtualBox NAT Network (LabNetwork, CIDR 192.168.100.0/24) with the gateway at 192.168.100.1.

Host / System	Lab Role	IP Address	Network Ports & Channels	Operating System
DESKTOP-US2NJE2	Monitored Windows Workstation	192.168.100.8	SplunkForwarder service, Sysmon64, SMB 445	Windows 11 Pro (64-bit)
wazuh-server	Central Splunk SIEM	192.168.100.7	9997/tcp (Ingestion), 8000/tcp (Web UI)	Ubuntu Server 24.04 LTS (Splunk 10.4.3)
kali	Adversary Node	192.168.100.6	Workstation: KALI, Port 2224	Kali Linux (Rolling)
Hypervisor Host	Analyst Access Gateway	192.168.100.1	Host Port 18000 → 8000 (Splunk Web)	Windows 11 Host

3. Visual Endpoint Verification & Forwarder Architecture

Endpoint verification confirmed active service states for both `Sysmon64` and `Sp1unkForwarder`, with network attachment to the isolated subnet:



```
Writing request stream... (Number of bytes written: 13986191)

Name      DisplayName      Status
-----
Sysmon64   Sysmon64         Running
VBoxService VirtualBox Guest Additions Service Running

PS C:\WINDOWS\system32> cd $env:USERPROFILE\Downloads
PS C:\Users\natto\Downloads>
PS C:\Users\natto\Downloads> Invoke-WebRequest `
>> -Uri "https://download.splunk.com/products/universalforwarder/releases/10.4.3/windows/splunkforwarder-10.4.3-4174a2deda5d-windows-x64.msi" `
>> -OutFile "splunkforwarder.msi"
```

Exhibit EX-01: Windows 11 verifying running Sysmon64 service and UF installer fetch.

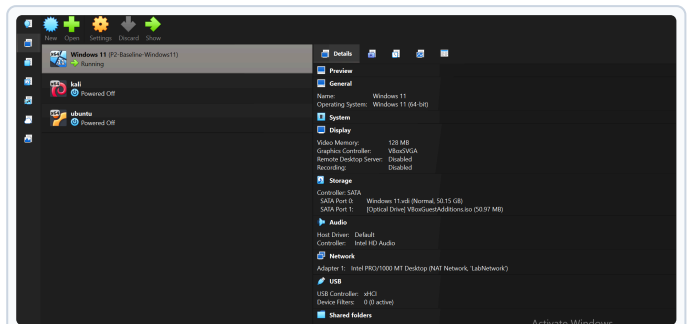


Exhibit EX-02: VirtualBox Manager confirming Windows 11 attachment to LabNetwork.

3.1. Sysmon Telemetry & XML Parsing Strategy

Sysmon events are collected via `inputs.conf` with `renderXml = true` to ensure structured XML capture. Because Splunk does not automatically parse `EventCode` from this raw XML format, detection queries utilize targeted regular expression parsing:

```
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=1
```

3.2. Forwarder Configuration Stanzas

The Universal Forwarder configuration on the Windows 11 endpoint segregates telemetry into separate indexes:

```
# C:\Program Files\SplunkUniversalForwarder\etc\apps\windows_ta\local\inputs.conf
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
checkpointInterval = 5
current_only = 0
disabled = 0
index = sysmon
renderXml = true

[WinEventLog://Security]
checkpointInterval = 5
current_only = 0
disabled = 0
index = windows

[WinEventLog://System]
checkpointInterval = 5
current_only = 0
disabled = 0
index = windows
```

4. MITRE ATT&CK Framework Mapping

Tactic	Technique ID	Technique Name	Detection Implementation in P2
Execution	T1059.001	PowerShell	Monitoring Sysmon Event ID 1 process launches and Script-Block Event ID 4104.
Execution	T1059.003	Windows Command Shell	Capturing <code>cmd.exe</code> spawns with full command-line arguments (Event ID 4688 / Sysmon 1).
Discovery	T1046	Network Service Discovery	Tracking Kali SMB reconnaissance targeting port 445 on the Windows endpoint.
Credential Access	T1110	Brute Force	Detection of failed logon attempts (Event ID 4625) originating from external IP <code>192.168.100.6</code> .
Defense Evasion	T1078.003	Valid Accounts: Local Accounts	Auditing successful network authentication (Event ID 4624) by remote workstation <code>KALI</code> .

5. Detection Engineering & SPL Rule Suite

The following SPL correlation queries form the core detection library for Windows endpoint surveillance and adversary tracking:

UC-01: Sysmon Process Creation Baseline (Event ID 1)

```
index=sysmon earliest=-24h
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=1
| stats count
```

Observed Telemetry: 3,994 process creations captured; establishes normal system execution profiles.

UC-02: Sysmon DNS Query Monitoring (Event ID 22)

```
index=sysmon earliest=-24h
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=22
| stats count
```

Observed Telemetry: 100 DNS resolution events cataloged; enables discovery of domain generation algorithms (DGA) and C2 beacons.

UC-03: Kali Attacker Authentication Failures (Event ID 4625)

```
index=windows "192.168.100.6" "<EventID>4625</EventID>"
| stats count
```

Observed Telemetry: Flagged SMB anonymous null-session connection attempts (`smbclient -L //192.168.100.8 -N`) rejected by Windows Security.

UC-04: Adversary Authentication Timeline Correlation

```
index=windows "192.168.100.6" ("<EventID>4625</EventID>" OR "<EventID>4624</EventID>")
| timechart count by EventID
```

Observed Telemetry: Correlated temporal distribution comparing 4625 failures with subsequent 4624 successful network logons.

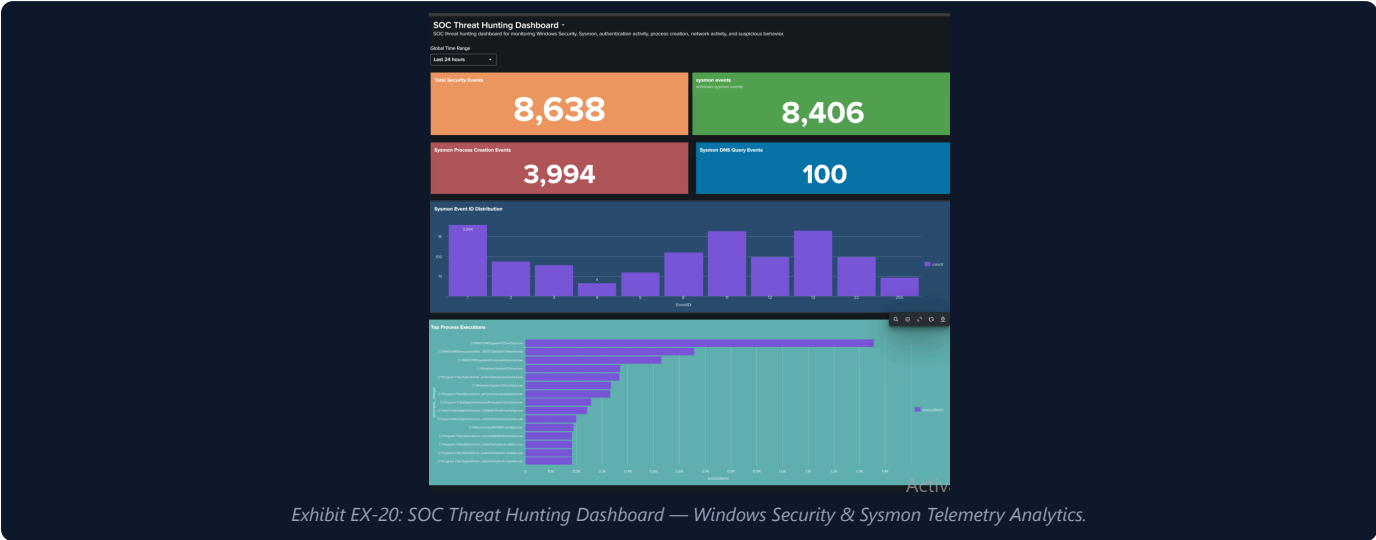
UC-05: Attacker Workstation & Account Attribution

```
index=windows "192.168.100.6"
| stats count by host, WorkstationName, TargetUserName
```

Observed Telemetry: Unambiguously identified remote workstation name `KALI` and IP `192.168.100.6` targeting victim `DESKTOP-US2NJE2`.

6. Operational SOC Dashboards Review

Project P2 provides two comprehensive Splunk dashboards delivering multi-layered situational awareness across general endpoint health and targeted adversary investigation.



6.1. Operational Panel Breakdown Across Dashboards

Dashboard	Panel Name	Visualization	SOC Analytical Function & Observed Metrics
Threat Hunting	Total Security Events	Single Value	8,638 events indexed across Windows Security event streams.
Threat Hunting	Sysmon Events	Single Value	8,406 events monitoring process, network, and registry operations.
Threat Hunting	Process Creation (ID 1)	Single Value	3,994 events tracking executable spawns on the Windows host.
Threat Hunting	DNS Queries (ID 22)	Single Value	100 events monitoring domain name resolutions from endpoint software.
Attacker Activity	Kali Attacker Events	Single Value (Red)	Total Windows Security events correlated with source IP 192.168.100.6.
Attacker Activity	Failed Auth (4625)	Single Value	Quantifies rejected authentication attempts during adversary reconnaissance.
Attacker Activity	Failed Logons by Host	Bar Chart	Visualizes targeted host: DESKTOP-US2NJE2.
Attacker Activity	Kali Workstation Name	Single Value	Extracts authenticated workstation string: KALI.
Attacker Activity	Network Logons (4624)	Single Value	Tracks authorized SMB protocol negotiation events.

7. Incident Investigation Case Studies

Scenario 1: SMB Enumeration & Anonymous Null-Session Probing

Incident Alert: The Kali Attacker Activity Dashboard registered an authentication failure (Event ID 4625) originating from 192.168.100.6 with workstation name KALI.

Forensic Investigation:

- Adversary Activity:** The attacker executed `smbclient -L //192.168.100.8 -N` to list Windows share permissions without providing valid credentials.
- Windows Defense:** Local Security Policy rejected null-session IPC\$ enumeration, generating Event ID 4625 with Status `0xC000006D` (Substatus `0xC0000064` — user does not exist).
- SOC Action:** Correlation confirmed reconnaissance activity; attacker was unable to map administrative shares (`C$`, `ADMIN$`).

Scenario 2: Authenticated Network Session Correlation

Incident Alert: Following enumeration, Event ID 4624 (Logon Type 3 — Network) was registered from the attacker IP.

Forensic Investigation:

- Logon Context:** The attacker initiated an authenticated SMB connection with valid lab credentials.
- Telemetry Corroboration:** Sysmon Event ID 3 confirmed an established TCP connection on port 445 from 192.168.100.6.
- SOC Action:** Session duration and process tree analysis confirmed no secondary payload was executed.

8. Endpoint Hardening & Defensive Recommendations

Defensive Domain	Mitigation Recommendation	Technical Implementation
SMB Hardening	Disable SMBv1, restrict SMB null sessions, enforce SMB signing.	<code>Set-SmbServerConfiguration -EnableSMB1Protocol \$false -RequireSecuritySignature \$true</code>
Audit Policy	Enable Process Creation with command-line auditing (Event ID 4688).	Configure GPO: <i>Include command line in process creation events</i> = Enabled.
PowerShell Security	Enable Script-Block Logging (4104) and Constrained Language Mode.	Enable <i>Turn on PowerShell Script Block Logging</i> in Administrative Templates.
Credential Protection	Enable Windows Defender Credential Guard and LSA Protection.	Configure registry: <code>RunAsPPL = 1</code> in <code>HKLM\SYSTEM\CurrentControlSet\Control\Lsa</code> .

9. Conclusion & Portfolio Impact

Project P2: Windows Security Monitoring with Splunk establishes an enterprise-grade Windows host monitoring and adversary investigation capability. The implementation validates mastery across Windows Advanced Auditing, Sysmon engineering, SPL detection query design, and incident reconstruction within a multi-node lab environment.